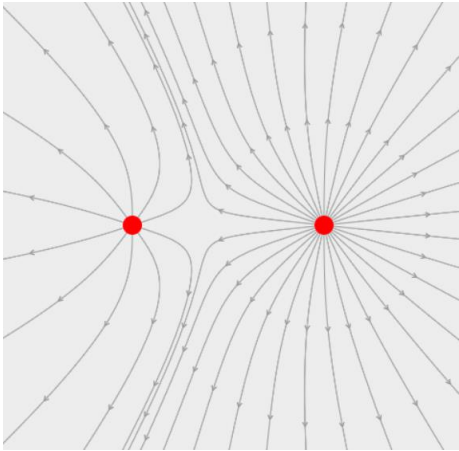


8(d)(i)	For gravitational potential at a point, the displacement of a small test mass from infinity to that point is always in the opposite direction to the external force acting on the small test mass because gravitational forces between two masses are always attractive. For electric potential at a point, the displacement of the small test charge from infinity to that point can be in the opposite or same direction to the external force acting on the small positive test charge because electric forces between two charges can be attractive or repulsive.	B1 B1
	OR [w.r.t. to DHS G.Field Tut] Due to the attractive nature of the gravitational force, work done by an external force to bring any mass from infinity to that point is always negative. However, electric force can be attractive or repulsive, thus work done by an external force to bring any charge from infinity to that point can be positive or negative.	B1 B1
CKW MC	Some candidates referenced the formulae for gravitational and electric potential in their discussions but failed to address the key conceptual difference between the two types of potentials, despite their similar definitions i.e. in terms of work done by an external force.	
8(d)(ii)	The electric force between an alpha particle and a gold nucleus is many orders of magnitude larger than the gravitational force between the alpha particle and gold nucleus.	B1

CKW MC	- A correct identification of the forces involved, and their comparison is needed to successfully answer this part. - No matter how small the gravitational force between an alpha particle and a gold nucleus is, if it is the only force present, then it cannot be neglected. - The order of magnitude of a gravitational force between two point masses is not only dependent on the product of the masses but also on the gravitational constant and the separation of the masses. - At the same separation r, the electric force is given by $F_e = \frac{(79e)(2e)}{4\pi\epsilon_0 r^2}$ while the gravitational force is given by $F_g = G \frac{(197u)(4u)}{r^2}$. Thus, $\frac{F_e}{F_g} = \frac{(79e)(2e)}{4\pi\epsilon_0 G(197u)(4u)}$$= \frac{(79)(2)(1.6 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(6.67 \times 10^{-11})(197)(4)(1.66 \times 10^{-27})^2}$$\approx 3 \times 10^{35}$	
8(d)(iii)	Assumption: The gold nucleus acts as a point positive charge. $V = \frac{Q}{4\pi\epsilon_0 r}$ $= \frac{79(1.6 \times 10^{-19})}{4\pi(8.85 \times 10^{-12})(2.6 \times 10^{-12})}$ $= 43714$ $= 44000 \text{ V}$	B1 C1 A1
CKW MC	- The assumption that “the gold nucleus is isolated” is not valid since the electric potential required is specified to be due to the gold nucleus in the question. - The formula used is only applicable for a point charge, which the gold nucleus is not physically in reality. - Some students thought the charge of a proton is 1 C.	

8(d)(iv)		B1: Direction of arrows B1: Position of neutral point (closer to alpha particle) B1: Field line pattern: • Field lines do not intersect. • Density of field lines around alpha particle is less than gold nucleus. • Symmetrical about the horizontal, asymmetrical about the vertical. • The circle represents an equipotential volume. The field lines are normal to the surface. [The suggested answer shows the simulated field lines between a charge of +1 C and +5 C. The actual field lines between a charge of +2e & +79e is more asymmetric.]
CKW MC	• Some candidates sketched ○ equipotential lines, or ○ magnetic field lines • instead of electric field lines. • The imaginary line joining the points of zero resultant electric field strength were observed to vertically straight in many cases; this is only possible if the charges have the same magnitude and same sign. Some sites to simulate electric field lines between two point charges: ○ E-Field Simulation - Ithaca College PER ○ Electric field line simulator	